

QUBO Formulation: Remote-Sensing Image Segmentation

Remote-Sensing Image Segmentation

Problem Statement. Given a remote-sensing image represented as a grid graph with vertex set V and edge set E , the task is to partition the pixels into two segments. Each pixel is assigned a label from a binary set, and the assignment must minimize a global segmentation energy. This energy comprises unary terms that encode the preference for assigning a specific label to a pixel, and pairwise terms that penalize boundary cuts between neighboring pixels with different labels. The goal is to determine the optimal binary labeling that minimizes this energy function.

Decision Variables. Let $N = |V|$ denote the total number of pixels in the image. For each pixel $i \in V$, we introduce a binary decision variable $x_i \in \{0, 1\}$. The variable x_i encodes the segment assignment for pixel i , where $x_i = 0$ corresponds to the first segment and $x_i = 1$ corresponds to the second segment. The complete solution is represented by the vector $x = (x_0, x_1, \dots, x_{N-1})^\top \in \{0, 1\}^N$.

QUBO Derivation. The segmentation energy is directly expressed as a quadratic unconstrained binary optimization (QUBO) objective. We derive the corresponding Hamiltonian $H(x)$ and its coefficient matrix Q through the following steps.

Step 1. The original objective function to be minimized is given by the sum of unary label costs and pairwise boundary penalties:

$$\min_{x \in \{0,1\}^N} H(x) = \sum_{i \in V} a_i x_i + \sum_{(i,j) \in E} b_{ij} x_i x_j, \quad (1)$$

where a_i denotes the unary cost coefficient for pixel i , and b_{ij} denotes the pairwise boundary penalty coefficient for the edge connecting pixels i and j .

Step 2. The problem formulation imposes no additional equality or inequality constraints beyond the binary domain restriction $x_i \in \{0, 1\}$ for all $i \in V$. Therefore, no constraint equations need to be written.

Step 3. Since there are no constraints, no penalty terms are required. The formulation relies solely on the binary domain $\{0, 1\}^N$, and the objective function is already in quadratic form. Consequently, the penalty contribution to the Hamiltonian is identically zero.

Step 4. Combining the objective and the absence of penalty terms, the single QUBO Hamiltonian $H(x)$ is:

$$H(x) = \sum_{i \in V} a_i x_i + \sum_{(i,j) \in E} b_{ij} x_i x_j. \quad (2)$$

Step 5. We read off the entries of the upper-triangular QUBO matrix Q and the constant offset c from the expanded Hamiltonian. For any $i, j \in V$, the matrix entries and offset are given in closed form as:

$$Q_{i,i} = a_i, \quad (3)$$

$$Q_{i,j} = b_{ij} \quad \text{for } (i,j) \in E \text{ with } i < j, \quad (4)$$

$$Q_{i,j} = 0 \quad \text{for } (i,j) \notin E \text{ with } i < j, \quad (5)$$

$$c = 0. \quad (6)$$

The optimization problem is thus equivalent to $\min_{x \in \{0,1\}^N} x^\top Q x + c$.

Penalty Weight Justification. In constrained QUBO formulations, a penalty weight λ is typically introduced to enforce hard constraints by adding squared violation terms to the objective. For this remote-sensing segmentation problem, the specification contains no constraints beyond the intrinsic binary domain of the variables. Consequently, no penalty weight λ is defined or required. The formulation is inherently unconstrained, and the binary restriction $x_i \in \{0,1\}$ is naturally satisfied by the hardware or solver domain without auxiliary penalty terms.

Correctness Argument. Minimizing $H(x)$ over the domain $\{0,1\}^N$ recovers the optimal segmentation because the Hamiltonian is constructed to be exactly equivalent to the physical segmentation energy. Every binary assignment $x \in \{0,1\}^N$ corresponds to a valid segmentation of the image pixels, meaning all feasible solutions are evaluated directly without penalty. Since the objective function is a faithful representation of the segmentation cost, the global minimum of $H(x)$ over the binary hypercube must correspond to the labeling that achieves the minimum possible segmentation energy. Thus, any solver returning the ground state of $H(x)$ yields the optimal partition and its associated minimum energy value.

Illustrative Example. Consider a concrete instance with $N = 3$ pixels, where the pixel graph is fully connected such that $E = \{(0,1), (1,2), (0,2)\}$. The instance parameters are the unary costs a_0, a_1, a_2 and the pairwise weights b_{01}, b_{12}, b_{02} . Instantiating the general formulas from the derivation yields the following concrete objective:

$$H(x) = a_0 x_0 + a_1 x_1 + a_2 x_2 + b_{01} x_0 x_1 + b_{12} x_1 x_2 + b_{02} x_0 x_2. \quad (7)$$

There are no penalty terms, as the problem remains unconstrained. The resulting QUBO matrix entries and offset are explicitly:

$$Q_{0,0} = a_0, \quad Q_{1,1} = a_1, \quad Q_{2,2} = a_2, \quad (8)$$

$$Q_{0,1} = b_{01}, \quad Q_{1,2} = b_{12}, \quad Q_{0,2} = b_{02}, \quad (9)$$

$$c = 0. \quad (10)$$

For this $N = 3$ instance, the binary search space contains $2^3 = 8$ possible assignments. The optimal solution x^* is the binary vector in $\{0, 1\}^3$ that minimizes the expression above, and the minimum energy is $H(x^*)$. Substituting numerical values for a_i and b_{ij} allows direct enumeration or quantum/classical solver execution to identify x^* and $H(x^*)$.